

Product Description

Product Information

Product name	Picosecond Pulsed Fiber-Coupled Laser
Product number	PLD-XXX-SM-FC-30-U
Manufacturer	FLIM LABS S.r.l.
Registered Address	Via della Farnesina, 3 – 00135 Rome (RM), Italy



General Description

The PLD is a solid-state diode laser source that can operate in both continuous (CW) and pulsed modes with different frequencies and power settings. The laser emits at a single wavelength, which can be selected at the time of purchase from a list of available model listed in Table 1. The laser device is therefore available as both a visible and invisible source. The maximum laser power output allows the product to be classified as Class 3R according to the technical standard IEC EN CEI 60825-1 ed.3 (2014).

The source is a self-contained unit, designed to operate on a desktop or bench and to be connected to a microscopy instrument or other imaging equipment via optical fibers.

The source is available in different models depending on the wavelength of the laser radiation as shown in Table 1.

Table 1: Available PLD laser models.

Model	Wavelength	Radiation Type
PLD-405-SM-FC-30-U	405 nm	Visible
PLD-445-SM-FC-30-U	445 nm	Visible
PLD-488-SM-FC-30-U	488 nm	Visible
PLD-520-SM-FC-30-U	520 nm	Visible
PLD-635-SM-FC-30-U	635 nm	Visible
PLD-808-SM-FC-30-U	808 nm	Invisible
PLD-850-SM-FC-30-U	850 nm	Invisible

The model number encodes the product specifications in the following scheme:

- SM stands for single-mode
- FC for fiber-receptacle
- 30 is the laser diode (LD) power in mW
- U indicates the USB power interface

The device incorporates a laser diode capable of emitting a maximum optical power of 30 mW (class 3B), but is equipped with a control system that limits the supply current so that, even under fault conditions, the maximum laser power deliverable makes the device a class 3R laser product.

The laser emits a single-mode Gaussian beam because it is propagated inside an optical fiber with a 4 μm core.

Intended Use and Misuse

Intended Use

The device described in this manual is designed for standalone use or incorporation into other equipment or machinery for research and diagnostic applications in the field of fluorescence analysis.

Reasonably Foreseeable Misuse

The following uses are not permitted:

- Use outside described applications
- Use with incompatible components
- Operation outside specified limits
- Unauthorized modification
- Use with incorrect power supply

Environmental and Operating Conditions

The device is designed for indoor laboratory use only. Operate the device only within the specified environmental limits:

- Operating temperature: 10°C to +40°C
- Storage temperature: -20°C to +70°C
- Relative humidity: up to 80% non-condensing
- Stable, vibration-free surface

Do not operate the device in explosive atmospheres or in the presence of flammable gases. Ensure adequate ambient lighting during installation and adjustment to avoid connection errors.